



Modelling Financial Markets

Chapter 4: Portfolio Optimization



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Acronyms used in this material

- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LO: Global Minimum-Variance portfolio, Long-Only
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit-Wolf covariance
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ LO: Long-Only, no short sales
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ MV: Mean–Variance tangency portfolio estimated from sample moments
- ▣ MV-LO: Mean–Variance tangency portfolio, Long-Only
- ▣ OOS: Out-Of-Sample
- ▣ US: United States

Outline

Motivation

Mean–Variance Optimisation

The Limits of Markowitz

Covariance Shrinkage

Black–Litterman

Risk Parity and Hierarchical Risk Parity

Out-of-Sample Evaluation

A Portfolio of BVB Blue Chips

Conclusions

References

What Is Portfolio Optimisation?

Definition

- ▣ A **portfolio** is a vector of weights $w = (w_1, \dots, w_N)^\top$, the shares of wealth invested in N assets, with $w^\top \mathbf{1} = 1$
- ▣ **Portfolio optimisation**: choose w to maximise a criterion (return per unit of risk, utility) subject to constraints

Key message of this chapter

- ▣ The optimal portfolio is easy to write and hard to estimate: judge every rule **out of sample**, after costs, with a test

Why It Matters

- ▣ **Asset managers and pension funds:** strategic allocation across equities, bonds, credit, gold
- ▣ **Private investors:** how much in local stocks, how much abroad?
- ▣ **Risk management:** the same covariance matrix drives portfolio risk (Chapters 6–7)
- ▣ **Econometrics:** N means and $N(N + 1)/2$ covariances estimated from short samples
- ▣ Textbook treatment: [Franke, Härdle & Hafner \(2019\)](#), Part I (portfolio theory)

The Central Tension

In theory

- ▣ Known means and covariances
- ▣ The mean–variance portfolio is optimal
- ▣ More assets always help

In practice

- ▣ Parameters are estimated with error
- ▣ The optimiser loads on the largest errors
- ▣ More assets, more parameters, more noise

- ▣ This chapter: better inputs (shrinkage, Black–Litterman), rules that ignore means (risk parity, HRP) and a fair out-of-sample comparison

Data Used in This Chapter

Universe	Assets	Months
Sectors	9 US sector ETFs (XLB ... XLY)	1999–2026
Multi-asset	SPY, EFA, EEM, TLT, IEF, LQD, HYG, GLD	2007–2026
Combined	9 sectors + 7 multi-asset ETFs (no SPY)	2007–2026
Romania	8 BVB blue chips, BET, BET-TR	2014–2026

- Monthly returns from month-end prices on common trading days; US data end in July 2026, BVB data in August 2026
- Risk-free rate: one-month Treasury bill, [Kenneth French Data Library](#)

The Assets

- ▣ Equities: SPY (S&P 500), EFA (developed markets outside North America), EEM (emerging markets)
- ▣ US Treasuries: TLT (maturities above 20 years), IEF (7–10 years)
- ▣ Credit: LQD (investment-grade corporate bonds), HYG (high-yield corporate bonds)
- ▣ Gold: GLD; sectors: XLK technology, XLU utilities, XLP consumer staples, . . .
- ▣ BVB: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL; BET is a price index, BET-TR reinvests dividends

Notation and Performance Measures

- Simple return $R_{i,t} = P_{i,t}/P_{i,t-1} - 1$; excess return $R_{i,t}^e = R_{i,t} - R_{f,t}$
- Portfolio: $R_{p,t} = \mathbf{w}^\top \mathbf{R}_t$, mean $\mu_p = \mathbf{w}^\top \boldsymbol{\mu}$, variance $\sigma_p^2 = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}$
- Annualisation of monthly figures: mean $\times 12$, volatility $\times \sqrt{12}$
- **Sharpe ratio** ([Sharpe \(1966\)](#)): excess return per unit of volatility

$$SR_p = \frac{\mathbb{E}[R_p^e]}{\sigma(R_p^e)}$$

- **Maximum drawdown**: largest fall of wealth from a previous peak, $\min_t (W_t / \max_{s \leq t} W_s - 1)$

The Markowitz Problem

- ▣ **Markowitz (1952)**: investors care about the mean and the variance of portfolio returns
- ▣ **Efficient frontier**: minimum variance for every target mean m

$$\min_w w^\top \Sigma w \quad \text{s.t.} \quad w^\top \mu = m, \quad w^\top \mathbf{1} = 1$$

- ▣ Equivalent utility form, risk aversion γ : $\max_w w^\top \mu - \frac{\gamma}{2} w^\top \Sigma w$
- ▣ Inputs: N means, N variances, $N(N-1)/2$ covariances
- ▣ Weights may be negative: a **short sale** (selling a borrowed asset)

Two Assets: Diversification

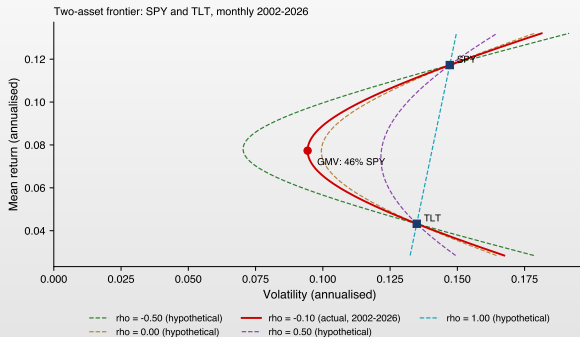
- Weight w in asset 1, $1 - w$ in asset 2, correlation ρ

$$\sigma_p^2 = w^2 \sigma_1^2 + (1 - w)^2 \sigma_2^2 + 2w(1 - w)\rho \sigma_1 \sigma_2$$

- $\rho = 1$: no diversification, σ_p is linear in w
- $\rho < 1$: σ_p lies below the straight line; $\rho = -1$: a riskless mix exists
- Minimum-variance weight (set $\partial \sigma_p^2 / \partial w = 0$)

$$w^* = \frac{\sigma_2^2 - \rho \sigma_1 \sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho \sigma_1 \sigma_2}$$

Two-Asset Frontiers: Stocks and Treasuries



□ SPY and TLT, monthly total returns, August 2002 – July 2026 ($T = 288$)

Reading the Two-Asset Frontier

- ▣ SPY: mean 11.7%, volatility 14.7%; TLT: mean 4.3%, volatility 13.5%
- ▣ Sample correlation $\rho = -0.10$: Treasuries hedged equities on average
- ▣ Minimum-variance mix: 46% SPY, volatility 9.4%, lower than either asset
- ▣ The curves for hypothetical ρ show that correlation, not volatility, drives diversification
- ▣ Caveat: the correlation changes over time: daily -0.40 in 2002–2021, $+0.08$ in 2022 (Chapter 6)

Worked Example: The Minimum-Variance Mix, Step by Step

- ▣ Inputs: SPY $\sigma_1 = 0.1472$, TLT $\sigma_2 = 0.1349$, $\rho = -0.1005$
- ▣ Numerator: $\sigma_2^2 - \rho\sigma_1\sigma_2 = 0.01820 + 0.00199 = 0.02019$
- ▣ Denominator: $\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 = 0.02166 + 0.01820 + 0.00399 = 0.04385$
- ▣ $w^* = 0.02019/0.04385 = 0.461$ in SPY, 0.539 in TLT
- ▣ $\sigma_p = 9.4\%$, mean 7.7%: two-thirds of the SPY mean with less than two-thirds of its risk

Tobin: Separation



James Tobin, Nobel Prize in Economics 1981

Photo: unknown author (1962); public domain;
[Wikimedia Commons](#)

- **Tobin (1958)**: add a riskless asset paying R_f
- Every investor holds the **same** risky portfolio, mixed with cash or borrowing
- Risk aversion changes only the mix, not the risky portfolio: **two-fund separation**
- The best risky portfolio maximises the Sharpe ratio: the **tangency portfolio**
- Separation is the bridge from Markowitz to the CAPM (Chapter 3)

Two Special Portfolios

Global minimum variance (GMV)

- Lowest variance of all portfolios

$$w_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}$$

- Needs **no** expected returns

- In this chapter, MV denotes the tangency rule estimated from sample means and covariances

Tangency (maximum Sharpe)

- Highest Sharpe ratio

$$w_{\text{MV}} = \frac{\Sigma^{-1} \mu}{\mathbf{1}^\top \Sigma^{-1} \mu}$$

- μ : expected **excess** returns

Merton: the Frontier in Closed Form



Robert C. Merton, Nobel Prize in
Economics 1997

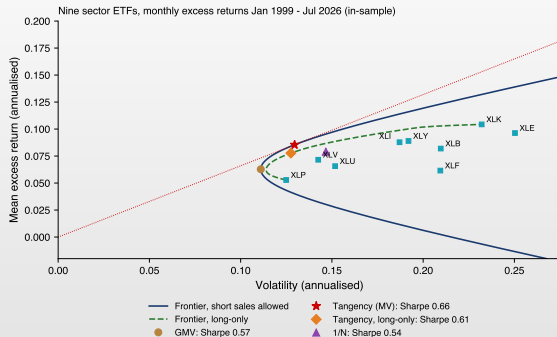
Photo: Digarnick (2006); public domain;
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- Merton (1972): with short sales, the frontier is a hyperbola
- Let $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, $B = \mathbf{1}^\top \Sigma^{-1} \boldsymbol{\mu}$, $C = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$, $D = AC - B^2$

$$\sigma^2(m) = \frac{Am^2 - 2Bm + C}{D}$$

- The GMV sits at $m = B/A$ with $\sigma^2 = 1/A$
- Every frontier portfolio is a mix of GMV and tangency

The Frontier of US Sectors



- Monthly excess returns, January 1999 – July 2026 ($T = 331$); all parameters estimated on the full sample

Reading the Sector Frontier

- ▣ Tangency: Sharpe 0.66; GMV: 0.57 (volatility 11.1%); $1/N$: 0.54
- ▣ Tangency weights: -0.37 financials, -0.21 materials, $+0.37$ health care
- ▣ Long-only tangency: Sharpe 0.61; materials, financials and industrials get exactly zero
 - ▶ A **corner solution**: the constraint binds and assets drop out
- ▣ The gain of the optimiser over $1/N$ is small even **in sample**, with perfect hindsight

Constraints

- ▣ **Long-only (LO)**: $w_i \geq 0$; no closed form, solved by quadratic programming
- ▣ Position limits: $w_i \leq 0.2$; sector or asset-class limits: $\sum_{i \in G} w_i \leq c_G$
- ▣ Leverage limit: $\sum_i |w_i| \leq L$
- ▣ **Frost & Savarino (1988)**: constraining weights improved out-of-sample performance
- ▣ In this chapter: MV-LO and GMV-LO are the long-only versions of MV and GMV

Recap: Mean–Variance Optimisation

- ▣ The frontier trades mean against variance; correlation drives diversification
- ▣ Two special portfolios: GMV, which needs no means, and tangency, which maximises the Sharpe ratio
- ▣ With a riskless asset, the risky part of every efficient portfolio is the tangency portfolio
- ▣ Long-only constraints create corner solutions
- ▣ Everything so far assumed the parameters are **known**

Means Are Hard to Estimate

- Standard error of an annualised mean over Y years: $\sigma/\sqrt{Y}$, whatever the sampling frequency
- Merton (1980): sampling more often sharpens variances, not means
- Covariances: the estimation error falls with the **number of observations**
- Chopra & Ziemba (1993): errors in means cost about ten times more than errors in variances
- Hence rules that need no means (GMV, risk parity) are often more robust

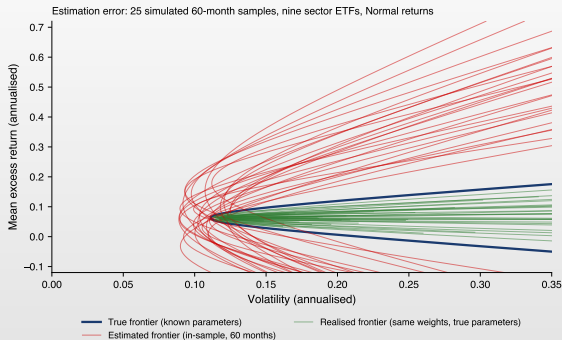
Worked Example: How Many Years for a Precise Mean?

- SPY, 24 years of monthly data: mean 11.7%, volatility 14.7%
- Standard error: $0.147/\sqrt{24} = 3.0\%$; 95% interval [5.8%, 17.6%]
- For a standard error of 1%: $Y = (0.147/0.01)^2 \approx 217$ years
- An optimiser treats a 3% estimation error as information
- Monthly or daily data do not help: only the length of the sample does

Error Maximisation

- Michaud (1989): mean–variance optimisers are **estimation-error maximisers**
 - ▶ They overweight assets with overestimated means and underestimated risk
- Best & Grauer (1991): tiny changes in a mean can move the weights of many assets
- In-sample performance is therefore **biased upwards**
- Experiment: simulate returns from known parameters, optimise, evaluate with the true parameters

Estimated versus Realised Frontiers



- True parameters: sector sample means and covariances, 1999–2026; returns drawn from the Normal distribution

Reading the Simulation

- Red: frontiers estimated from 60 months, far too optimistic
- Green: the same portfolios evaluated with the true parameters, all **below** the true frontier
- 2000 simulated tangency portfolios, 60 months each
 - ▶ Median in-sample Sharpe 1.57; median true Sharpe 0.30; true optimum 0.66
 - ▶ 96% have a true Sharpe below $1/N$ (0.54); 7.5% below zero
- Even with Normal, stable returns, estimation error destroys the optimum

Unstable Weights



- Nine sector ETFs; each month the weights use only the previous 60 months (top panel clipped at ± 6)

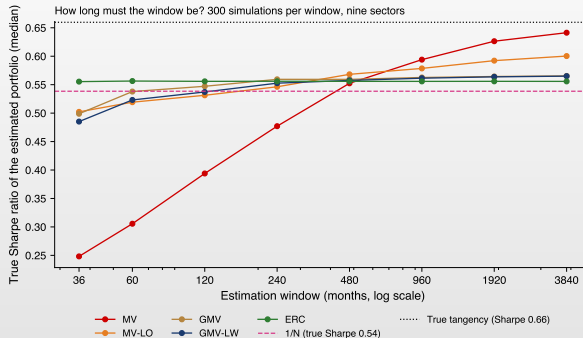
Reading the Rolling Weights

- MV: median gross exposure $\sum_i |w_i| = 4.9$; largest single weight 26 in absolute value
- Gross exposure above 3 in 90% of the months
- Explosions in 2004–2006 and 2009: $1^\top \Sigma^{-1} \hat{\mu}$ close to zero in the denominator
- GMV with shrinkage (next section): gross exposure 1.5, largest weight 0.94
- Unstable weights mean high turnover, hence high trading costs

The 1/N Benchmark

- ▣ **1/N**: equal weights $w_i = 1/N$, rebalanced each month; no estimation at all
- ▣ DeMiguel, Garlappi & Uppal (2009): 14 optimisation models on 7 data sets
 - ▶ None beats 1/N consistently in Sharpe ratio, certainty-equivalent return or turnover
- ▣ Their calibration: about 3000 months of data for sample MV to beat 1/N with 25 assets
- ▣ Kan & Zhou (2007): combining tangency and GMV reduces estimation risk
- ▣ Question: how long must the window be for our nine sectors?

How Long Must the Window Be?



- 300 simulated samples per window length; true parameters as in the previous simulation

Reading the Window-Length Experiment

- ▣ MV beats $1/N$ only with 480 months (40 years) of data: median true Sharpe 0.55 vs 0.54
- ▣ With 60 months, MV reaches 0.31, less than 60% of $1/N$
- ▣ Long-only MV, GMV and GMV with shrinkage are close to $1/N$ from 60 months on
- ▣ ERC (Section 6) is almost insensitive to the window: 0.56 throughout
- ▣ Here $1/N$ is nearly efficient: the true tangency Sharpe is only 0.66

Recap: The Limits of Markowitz

- ▣ Means are estimated with large errors; the optimiser amplifies them
- ▣ In-sample Sharpe ratios of optimised portfolios are biased upwards
- ▣ Sample MV weights are unstable and extreme
- ▣ $1/N$ is a hard benchmark: MV needs decades of data to beat it
- ▣ Remedies: better covariance estimates, better means, or rules that avoid means

Why the Sample Covariance Matrix Fails

- ▣ N assets: $N(N + 1)/2$ variances and covariances; 16 assets: 136 parameters from 60 months
- ▣ Sample eigenvalues are too dispersed: the largest too large, the smallest too small
- ▣ **Condition number**: largest over smallest eigenvalue; a large value means Σ^{-1} amplifies noise
- ▣ Optimisers invert Σ , so they load on the directions with the smallest estimated variance
- ▣ If $T < N$, the sample matrix is singular and cannot be inverted

The Shrinkage Estimator

- Shrink the sample matrix S towards a structured target F

$$\hat{\Sigma}_{\text{LW}} = \delta F + (1 - \delta) S, \quad 0 \leq \delta \leq 1$$

- Targets

- ▶ Single-index (market) model: Ledoit & Wolf (2003)
- ▶ Scaled identity: Ledoit & Wolf (2004a)
- ▶ Constant correlation, $f_{ij} = \bar{r}\sqrt{s_{ii}s_{jj}}$: Ledoit & Wolf (2004b), used here

- δ trades the variance of S against the bias of F

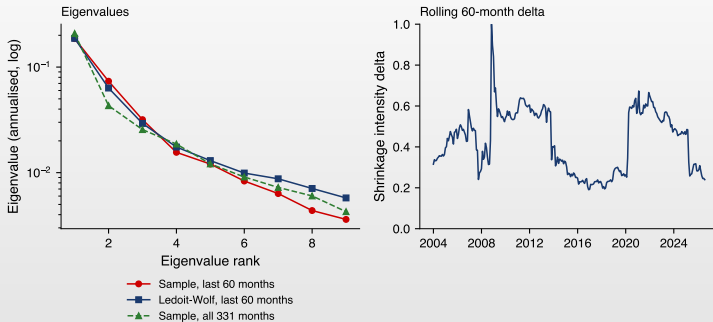
The Optimal Shrinkage Intensity

- Ledoit–Wolf choose δ to minimise the expected squared (Frobenius) distance to the true Σ

$$\delta^* = \max \left\{ 0, \min \left\{ 1, \frac{\hat{\kappa}}{T} \right\} \right\}, \quad \hat{\kappa} = \frac{\hat{\pi} - \hat{\rho}}{\hat{\gamma}}$$

- $\hat{\pi}$: sum of the asymptotic variances of the entries of S (noise)
- $\hat{\rho}$: their covariances with the target; $\hat{\gamma} = \|F - S\|^2$ (misspecification)
- Noisy data or a good target: large δ ; long samples: $\delta \rightarrow 0$

Shrinkage on US Sectors



- Nine sector ETFs, monthly excess returns; last window August 2021 – July 2026

Reading the Shrinkage Chart

- ▣ Last window: $\delta = 0.24$; the smallest eigenvalues are pulled up, the largest almost unchanged
- ▣ Condition number: sample 51.6, shrunk 32.5
- ▣ Rolling δ : mean 0.43, range 0.19–1.00
 - ▶ $\delta = 1$ in the early windows: 60 months are too few to beat the target
- ▣ GMV-LW: the GMV portfolio computed with $\hat{\Sigma}_{LW}$

Constraints Are Shrinkage Too

- Jagannathan & Ma (2003): the long-only GMV equals the unconstrained GMV of a modified covariance matrix
 - ▶ The constraint shrinks the covariances of the assets it binds on
- Imposing the “wrong” constraint (no short sales) can reduce out-of-sample risk
- The effect is largest when covariances are estimated from few observations
- Practical rule: shrink, constrain, or both, but never invert a raw sample matrix of many assets

Recap: Covariance Shrinkage

- The sample covariance matrix is noisy and ill-conditioned when N is large relative to T
- Shrinkage mixes it with a structured target; the data choose the weight δ
- The target matters: constant correlation suits one asset class better than a mix of stocks and bonds
- No-short constraints act as implicit shrinkage

Black and Litterman: Starting from Equilibrium



Fischer Black (1938–1995)

Photo: Dalmatine at English Wikipedia; public domain; Wikimedia Commons

- Black & Litterman (1992), developed at Goldman Sachs
- Problem: sample means give extreme, unintuitive weights
- Idea: start from the returns that make a **benchmark** portfolio optimal
- Add the investor's **views**, weighted by their confidence
- Result: weights that move away from the benchmark only where views exist

Step 1: Reverse Optimisation

- A mean–variance investor with risk aversion δ_{RA} holds $w^* = (\delta_{RA}\Sigma)^{-1}\mu$
- Reverse it: the returns that make the benchmark w_b optimal

$$\pi = \delta_{RA} \Sigma w_b$$

- π : **implied** (equilibrium) excess returns; with market weights, the CAPM returns
- Here: no market capitalisations for the sectors, so $w_b = 1/N$, $\delta_{RA} = 2.5$

Step 2: Views and the Posterior

▣ K views: $P\boldsymbol{\mu} = \mathbf{q} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N(0, \boldsymbol{\Omega})$; prior $\boldsymbol{\mu} \sim N(\boldsymbol{\pi}, \tau\boldsymbol{\Sigma})$

▣ Posterior mean: a precision-weighted average

$$\boldsymbol{\mu}_{\text{BL}} = [(\tau\boldsymbol{\Sigma})^{-1} + P^\top \boldsymbol{\Omega}^{-1} P]^{-1} [(\tau\boldsymbol{\Sigma})^{-1} \boldsymbol{\pi} + P^\top \boldsymbol{\Omega}^{-1} \mathbf{q}]$$

▣ Weights: $w_{\text{BL}} = (\delta_{\text{RA}} \boldsymbol{\Sigma})^{-1} \boldsymbol{\mu}_{\text{BL}}$

▣ He & Litterman (2002): $\tau = 0.05$, $\boldsymbol{\Omega} = \text{diag}(P \tau \boldsymbol{\Sigma} P^\top)$; then only the assets in the views move

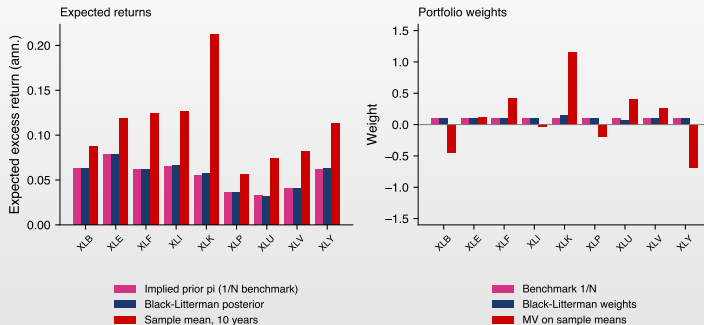
Worked Example: One Asset, One View

- Prior $\pi = 5\%$, $\sigma = 16\%$, $\tau = 0.05$: prior variance $\tau\sigma^2 = 0.00128$
- View $q = 8\%$ with standard deviation 2% : $\omega = 0.0004$
- Precisions: $1/0.00128 = 781$, $1/0.0004 = 2500$

$$\mu_{\text{BL}} = \frac{781 \times 5\% + 2500 \times 8\%}{781 + 2500} = 7.29\%$$

- The view gets weight 0.76 because it is more precise than the prior
- Halve the confidence (standard deviation 4%) and the weight falls to 0.44

Black-Litterman on US Sectors



□ Σ from monthly excess returns, August 2016 – July 2026; one view: XLK minus XLU = 3% a year

Reading Black–Litterman

- ▣ Implied XLK minus XLU: 2.2%; view 3%; posterior 2.6% (the average, since Ω equals the prior variance of the view)
- ▣ Weights: XLK from 11.1% to 14.4%, XLU from 11.1% to 7.8%, all others unchanged
- ▣ Sample means instead: XLK 21.3% a year; MV weights +115% XLK, −68% XLY, −46% XLB
- ▣ Black–Litterman is Bayesian shrinkage of the means towards equilibrium

Recap: Black–Litterman

- The prior comes from a benchmark, not from noisy sample means
- Views enter with explicit confidence; weights move only where views exist
- Results depend on the benchmark, τ , Ω and the risk aversion
- Without views, Black–Litterman returns the benchmark itself

Risk Contributions

- ▣ Volatility is homogeneous of degree one in the weights (Euler)

$$\sigma_p = \sum_{i=1}^N w_i \frac{\partial \sigma_p}{\partial w_i} = \sum_{i=1}^N \frac{w_i (\Sigma w)_i}{\sigma_p}$$

- ▣ **Risk contribution** of asset i : $RC_i = w_i (\Sigma w)_i / \sigma_p$; percentage share RC_i / σ_p
- ▣ Equal weights are not equal risks: volatile assets dominate
- ▣ Risk budgeting: choose w so that the shares RC_i / σ_p equal preset budgets b_i

Equal Risk Contribution

- ERC (Maillard, Roncalli & Teïletche (2010)): $RC_i = RC_j$ for all i, j
- Computed from a convex problem, then normalised to sum to one

$$\min_{y>0} \frac{1}{2} y^T \Sigma y - \frac{1}{N} \sum_i \ln y_i, \quad w = y / 1^T y$$

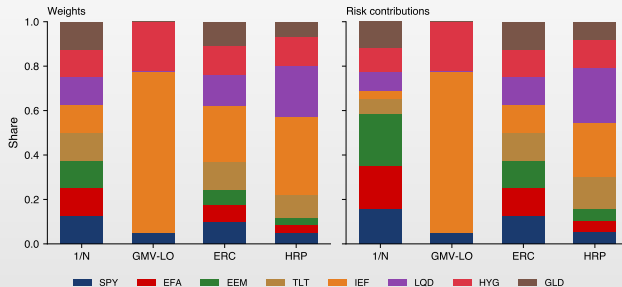
- Volatility ordering: $\sigma_{\text{GMV}} \leq \sigma_{\text{ERC}} \leq \sigma_{1/N}$
- Two assets: ERC is **inverse volatility**, $w_1 = \sigma_1^{-1} / (\sigma_1^{-1} + \sigma_2^{-1})$, for any ρ
- Needs no expected returns; only Σ

Worked Example: ERC with Stocks and Treasuries

- ▣ SPY volatility 14.7%, IEF volatility 6.6% (monthly, 2002–2026), $\rho = -0.10$
- ▣ ERC weight in SPY: $\frac{1/0.147}{1/0.147+1/0.066} = 0.31$
- ▣ Check: $w_1(\Sigma w)_1 = w_2(\Sigma w)_2$, each half of the variance
- ▣ A 60/40 portfolio instead: SPY carries 94% of the risk
- ▣ Risk parity funds lever the ERC portfolio to reach a target volatility

Weights versus Risk Contributions

Eight multi-asset ETFs, monthly May 2007 - Jul 2026



- Eight multi-asset ETFs, full-sample covariance matrix, May 2007 – July 2026 ($T = 231$)

Reading the Risk Contributions

- ▣ 1/N: equities and high yield receive half of the weight but 69% of the risk
- ▣ ERC: 12.5% of the risk from every asset; IEF gets the largest weight (25%)
- ▣ Long-only GMV: 73% IEF, 22% HYG, 5% SPY; five assets at zero
- ▣ Volatility: 1/N 9.4%, ERC 7.9%, HRP 7.1%, long-only GMV 5.8%
- ▣ ERC lies between 1/N and GMV, as the theory predicts

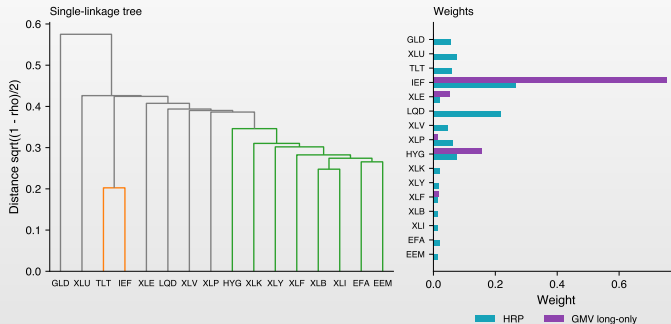
Hierarchical Clustering of Assets

- ▣ Correlation distance: $d_{ij} = \sqrt{(1 - \rho_{ij})/2} \in [0, 1]$
- ▣ **Single linkage**: repeatedly merge the two closest clusters; cluster distance = closest pair
- ▣ The result is a tree (**dendrogram**); cutting it gives groups of similar assets
- ▣ Mantegna (1999): the same distance and the **minimum spanning tree** reveal the hierarchy of a stock market
- ▣ Textbook: Härdle & Simar (2019), cluster analysis

Hierarchical Risk Parity

- HRP (López de Prado (2016)), three steps
 - ▶ Tree clustering on d_{ij} (single linkage)
 - ▶ Quasi-diagonalisation: reorder assets so that similar ones are adjacent
 - ▶ Recursive bisection: split the ordered list in two; split the weight in inverse proportion to the variance of each half
- Variance of a half: inverse-variance weights inside it, $\tilde{w}^\top \Sigma \tilde{w}$
- No matrix inversion: works even when Σ is singular
- All weights are positive

HRP on 16 ETFs



- Nine sectors and seven multi-asset ETFs, monthly, May 2007 – July 2026

Reading the Tree

- ▣ Closest pair: TLT and IEF (distance 0.20); gold joins last (0.57)
- ▣ Equity cluster: cyclical sectors, EFA, EEM and high-yield credit
- ▣ Defensive sectors (utilities, staples, health care) and energy join the equity cluster late
- ▣ HRP: all 16 weights positive, largest IEF 26.8% and LQD 21.9%
- ▣ Long-only GMV: 75.6% in IEF and only five assets above zero

Other Risk-Based Rules

- ▣ **Inverse volatility:** $w_i \propto 1/\sigma_i$; equals ERC when all correlations are equal
- ▣ **Maximum diversification** (Choueifaty & Coignard (2008)): maximise $w^\top \sigma / \sqrt{w^\top \Sigma w}$
- ▣ **Minimum variance:** GMV, the extreme risk-based rule
- ▣ All ignore expected returns: robust to estimation error, but optimal only under special assumptions
 - ▶ ERC is the tangency portfolio if all assets have the same Sharpe ratio and equal correlations

Recap: Risk-Based Allocation

- ▣ Risk contributions decompose portfolio volatility additively
- ▣ ERC equalises them; it lies between $1/N$ and GMV
- ▣ HRP uses a correlation tree and needs no matrix inversion
- ▣ These rules avoid expected returns entirely: less estimation error, no view on returns

Backtest Design

- ▣ **Out-of-sample** (OOS): weights at month t use only data up to $t - 1$
- ▣ Rolling window of 60 months; rebalancing every month
- ▣ Eight rules: 1/N, MV, MV-LO, GMV, GMV-LO, GMV-LW, ERC, HRP
- ▣ Three universes: sectors (OOS January 2004 – July 2026, 271 months), multi-asset and combined (May 2012 – July 2026, 171 months)
- ▣ Performance of excess returns: annualised mean, volatility, Sharpe ratio, maximum drawdown

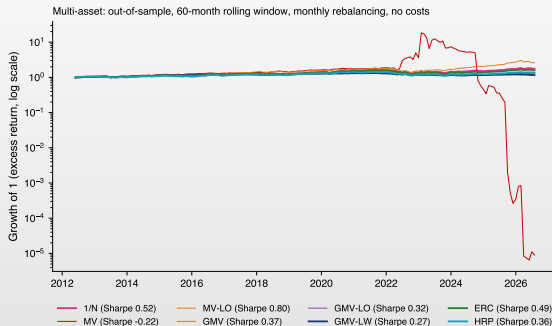
Turnover and Transaction Costs

- Between rebalancings weights drift with returns

$$w_{i,t}^+ = \frac{w_{i,t}(1 + R_{i,t})}{\sum_j w_{j,t}(1 + R_{j,t})}$$

- Turnover:** traded fraction of wealth, $TO_{t+1} = \sum_i |w_{i,t+1} - w_{i,t}^+|$
- Net return with cost c per unit traded: $R_{p,t}^{\text{net}} = R_{p,t} - c TO_t$
- 1/N also trades: it sells winners and buys losers every month

Out-of-Sample Growth: Multi-Asset



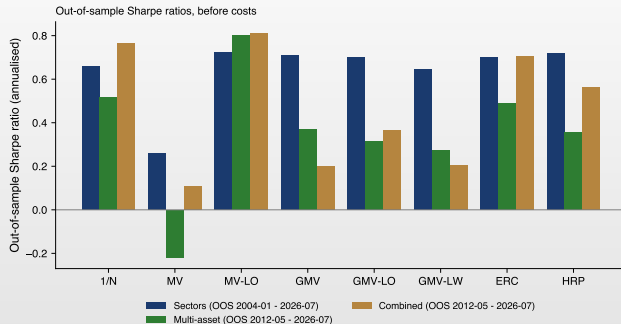
- Eight multi-asset ETFs; wealth of 1 invested in the excess returns, before costs

MFM_ch4_oos_backtest

Reading the Multi-Asset Backtest

- ▣ MV collapses: annualised volatility 518%, turnover 44 times wealth per month
- ▣ MV-LO has the highest Sharpe (0.80); 1/N 0.52; ERC 0.49
- ▣ GMV rules have the lowest volatility (3.8%–5.4%) but also the lowest Sharpe (0.27–0.37)
 - ▶ They load on Treasuries, whose excess returns were slightly negative after May 2012 (TLT -0.5% , IEF -0.2% a year)
- ▣ Is MV-LO really better, or lucky? A test is needed

Sharpe Ratios Across Universes



60-month rolling window, monthly rebalancing, annualised, before costs

MFM_ch4_oos_backtest

Results: Sector Universe

Rule	Mean	Volatility	Sharpe	Turnover	Max. drawdown
1/N	9.4%	14.3%	0.66	0.02	−50%
MV	17.8%	69.0%	0.26	3.86	−100%
MV-LO	10.7%	14.7%	0.72	0.23	−44%
GMV	8.5%	11.9%	0.71	0.23	−34%
GMV-LO	8.1%	11.5%	0.70	0.09	−36%
GMV-LW	7.6%	11.8%	0.65	0.14	−32%
ERC	9.3%	13.2%	0.70	0.03	−48%
HRP	9.0%	12.5%	0.72	0.08	−45%

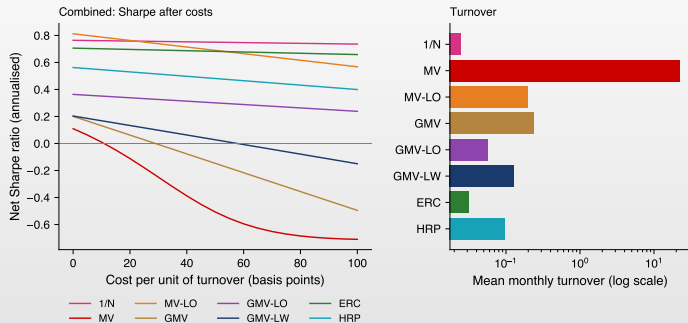
- January 2004 – July 2026; turnover: mean monthly traded fraction of wealth
- All rules except MV land within 0.07 of 1/N

Results: Combined Universe (16 ETFs)

Rule	Mean	Volatility	Sharpe	Turnover	Sharpe, 50 bp
1/N	7.9%	10.3%	0.76	0.02	0.75
MV	28.7%	261.8%	0.11	22.10	-0.51
MV-LO	8.0%	9.8%	0.81	0.20	0.69
GMV	0.8%	4.1%	0.20	0.24	-0.15
GMV-LO	2.0%	5.4%	0.36	0.06	0.30
GMV-LW	0.9%	4.4%	0.20	0.13	0.03
ERC	5.5%	7.8%	0.71	0.03	0.68
HRP	3.9%	7.0%	0.56	0.10	0.48

- May 2012 – July 2026; last column: net of 50 basis points per unit of turnover
- 1 basis point (bp) = 0.01%

Sharpe Ratios after Costs



- Combined universe (16 ETFs); cost c from 0 to 100 bp per unit of turnover

Reading the Cost Chart

- ▣ Turnover ranges from 0.02 (1/N) to 22 (MV) per month
- ▣ Net Sharpe turns negative at about 11 bp for MV, 29 bp for GMV, 58 bp for GMV-LW
- ▣ MV-LO overtakes 1/N only for costs below about 22 bp (Seminar, B8)
- ▣ 1/N and ERC barely move: low turnover is a hidden advantage
- ▣ Rule: report net performance for realistic costs, not only gross

Testing a Difference in Sharpe Ratios

- H_0 : $SR_1 = SR_2$ for two strategies evaluated on the same months
- Jobson & Korkie (1981): a test under i.i.d. Normal returns
- Ledoit & Wolf (2008): a robust version
 - ▶ $\hat{\Delta} = f(\hat{v})$, $v = (\mu_1, \mu_2, \mathbb{E}[r_1^2], \mathbb{E}[r_2^2])$, $f = \frac{\mu_1}{\sqrt{\mathbb{E}r_1^2 - \mu_1^2}} - \frac{\mu_2}{\sqrt{\mathbb{E}r_2^2 - \mu_2^2}}$
 - ▶ Standard error by the delta method: $\sqrt{\nabla f^\top \hat{\Psi} \nabla f / T}$, $\hat{\Psi}$ a HAC covariance
- Strategies on the same assets are highly correlated: small differences can be significant

The Block Bootstrap

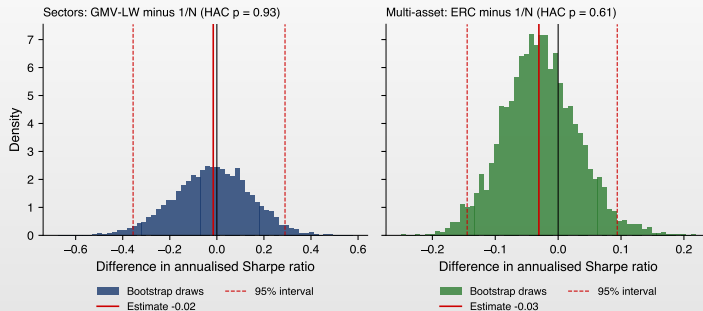


Bradley Efron, inventor of the bootstrap
(1979)

Photo: Ryan K. Morris (2007); public
domain; Wikimedia Commons

- Bootstrap: resample the data to approximate the sampling distribution of $\hat{\Delta}$
- Returns are dependent (volatility clustering): resample **blocks** of consecutive months (Künsch (1989))
- Circular version (as in Ledoit & Wolf (2008)): blocks wrap around the end of the sample
- Here: blocks of 6 months, 5000 draws; pairs of strategy returns resampled together
- p -value: share of draws with $|\hat{\Delta}^* - \hat{\Delta}| \geq |\hat{\Delta}|$

Is Anything Better than $1/N$?



- Bootstrap distributions of the Sharpe ratio difference against $1/N$, annualised

Reading the Tests

- ▣ Sectors, GMV-LW minus 1/N: -0.02 , 95% interval $[-0.36, 0.29]$, $p = 0.93$
- ▣ Multi-asset, ERC minus 1/N: -0.03 , interval $[-0.15, 0.09]$, $p = 0.59$
 - ▶ ERC and 1/N are so correlated that the interval is narrow, yet it still covers zero
- ▣ Sectors: no rule differs from 1/N at 5%; the closest is ERC, $+0.04$ ($p = 0.09$)
- ▣ Multi-asset: MV-LO $+0.29$ is **not** significant ($p = 0.12$)

Significant Differences: Mostly Losses

Universe	Rule minus 1/N	$\hat{\Delta}$	HAC p	Bootstrap p
Multi-asset	MV	-0.74	0.01	0.06
Multi-asset	HRP	-0.16	0.07	0.07
Combined	GMV	-0.57	0.03	0.03
Combined	GMV-LO	-0.40	0.05	0.05
Combined	GMV-LW	-0.56	0.02	0.03
Combined	MV-LO	+0.05	0.82	0.83

- No rule beats 1/N significantly in any universe
- GMV rules lose significantly on the combined universe: low risk, but too little return
- 21 tests: a few p -values near 0.05 are expected by chance (Chapter 3, multiple testing)

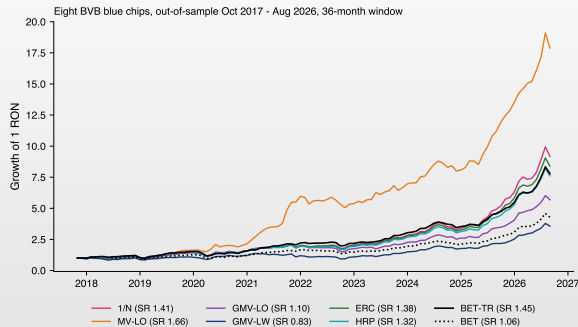
Recap: Out-of-Sample Evaluation

- Only out-of-sample, net-of-cost results are informative
- Sample MV fails in every universe; constraints and shrinkage tame it
- Risk-based rules and $1/N$ are hard to separate statistically
- Test Sharpe differences with HAC errors or a block bootstrap, and count your tests

Setup: a Romanian Investor

- ▣ Eight blue chips listed since 2014: TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL
- ▣ Monthly returns in RON, October 2014 – August 2026; 36-month window, first portfolio October 2017
- ▣ Rules: 1/N, MV-LO, GMV-LO, GMV-LW, ERC, HRP; benchmarks BET-TR and BET
- ▣ Sharpe ratios with $R_f = 0$ (no RON money-market series in the data): return per unit of risk, higher than excess-return Sharpe ratios
- ▣ Earlier Romanian evidence on diversification: Dragotă, Șerbănescu & Pele (2008)

BVB Portfolios versus BET-TR



□ Out-of-sample, October 2017 – August 2026 (107 months), before costs

Reading the BVB Backtest

- ▣ BET-TR: mean 24.7% a year, volatility 17.0%, Sharpe 1.45; BET: mean 17.8%, Sharpe 1.07
 - ▶ The gap of about 7% a year is the dividends: compare with BET-TR, never with BET
- ▣ 1/N 1.41, ERC 1.38, HRP 1.32: close to BET-TR
- ▣ MV-LO 1.66 (difference +0.21, $p = 0.44$); GMV-LW 0.83 (-0.63 , $p = 0.05$)
- ▣ Average correlation of the eight stocks: 0.44; the latest GMV-LO puts 70% in TLV

Beyond Equities: Digital Assets

- ▣ [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#): cryptocurrencies in portfolio allocation rules; gains depend on the rule and on liquidity
- ▣ [Pele et al. \(2023\)](#): are cryptos becoming alternative assets?
- ▣ Bitcoin: annualised volatility 172% (monthly, 2010–2026), IEF 6.6%
- ▣ Risk contributions grow with volatility: risk-based rules give Bitcoin a small weight automatically
- ▣ Robust mean–variance for crypto-assets: [Pele \(2023\)](#); Seminar 4, B5

Recap: BVB Portfolios

- Benchmark choice matters: BET omits about 7% a year of dividends
- No optimised BVB-only rule beats BET-TR significantly
- Eight stocks with average correlation 0.44: little room for diversification inside the BVB
- Open question: does diversification **beyond** BET pay? (Seminar 4, Part C)

Conclusions

- ▣ Markowitz is the right question with fragile answers: estimation error dominates
- ▣ Sample MV needs decades of data to beat $1/N$; out of sample it fails in every universe
- ▣ Shrinkage, constraints, Black–Litterman and risk-based rules make portfolios stable
- ▣ No rule beats $1/N$ significantly here; several lose after costs
- ▣ Evaluate out of sample, net of costs, with a test of the Sharpe difference

Links to Other Chapters

- ▣ **Chapter 3** (factor models): the tangency portfolio and the CAPM; factor covariance models
- ▣ **Chapters 5–6**: time-varying volatility and correlation for dynamic weights
- ▣ **Chapter 7**: portfolio Value-at-Risk and Expected Shortfall
- ▣ **Chapter 8**: backtesting without overfitting
- ▣ **Chapter 13**: machine learning for returns and portfolios

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Hierarchical Clustering
- ▣  Minimum Spanning Tree
- ▣  Multivariate Statistical Analysis

Code and data

- ▣ All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_04
- ▣ Risk-free rate: Kenneth French Data Library

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